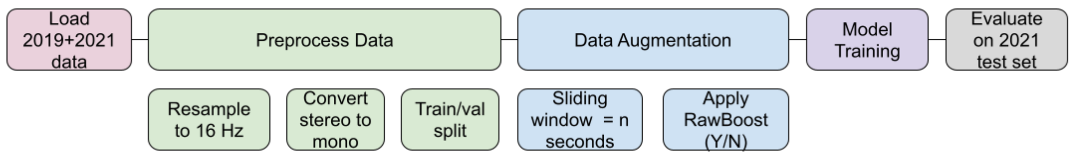
Mohini Gupta Model Update

<u>Model:</u> **W2V2**

<u>Pipeline:</u>



<u>Hyperparameters:</u>

Parameter	**Value**
Learning Rate	1e-5
Batch Size	16
Epochs	5
Optimizer	AdamW
Loss	CrossEntropy
Sample Rate	16 kHz
Sliding Window Length	Variable

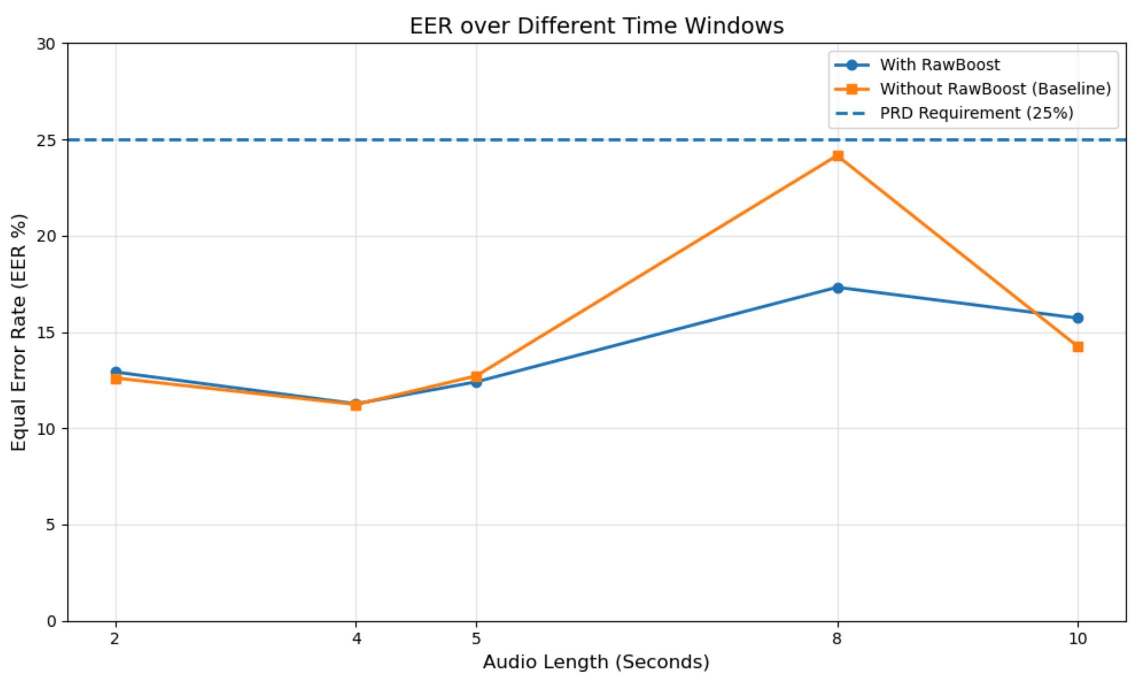
<u>Computational Cost:</u>

		Rawboost	
Len (s)		Yes	No (baseline)
	2s	4m/epoch 1hr test	N/A (trained/ tested on CPU, so time is much longer)
	4s	5min/epoch 1.5hr test	6min/epoch 1.5hr test
	5s	15 m/epoch 5hr test	14 m/epoch 5hr test
	8s	23 m/epoch 7.5hr test	24 m/epoch 7.5hr test
	10s	Tbd (still running)	30m/epoch 9.5hr test

Model Experiments:

Evaluations run on 2021 DF evaluation dataset

		Rawboost	
		Yes	No (baseline)
Len (s)	2s	12.92%	12.61%
	4s	11.27%	11.23%
	5s	12.41%	12.72%
	8s	17.32%	24.16%
	10s	15.72%	14.25%



Observations:
- 4s window has the lowest EER
  - Appears to be the sweet spot. Shorter windows don't have enough context, and larger ones start to pick up speaker characteristics rather spoofing artifacts
  - Reasoning:
    - It has the best balance between capturing spoofing artifacts and minimizing speech variability
- Models without rawboost are performing better by a very small margin for all windows but the 5s & 8s
  - Reasoning
    - Data is already well-conditioned/lacks environmental variations, so for this model (at all windows but 5s & 8s), rawboosting doesn't have a positive effect/much of an effect at all/the dataset does not benefit from robustness focused augmentation for this model
    - For 5s & 8s, the augmentation is acting as a regularizer
    - At 5s & 8s, the augmentation could also be preventing the model from overfitting to speaker characteristics by forcing it to learn more robust characteristics
- Performance degrades as the window increases
  - Longer context doesn't improve classification
  - The current model/architecture can't take advantage of extended temporal information
  - Potential Reasoning:
    - Could be due to information dilation and increased speaker variability
    - The EER for 10s could be lower than the one for 8s because the extra seconds contain enough additional speech information to regain cues lost from mean pooling at 8s
    - For longer utterances, speaker identity and linguistic content begin to dominate rather than spoofing cues
- Computational cost increases as window length increases, and remains about the same with or without rawboost

Next steps:
- Tune hyperparameters to ensure that signal-dependent and independent noises aren't overpowering the model's ability to detect bonafide vs deepfake for the rawboost models
- Try one other method to reduce degradation from Mustufa's email
- Send .pth files for best models to Hannah for explainability metrics/implementations
- Why does performance collapse after 5s?Do further analysis on why 8s and 10s perform much worse. Measure/remove silence in clips
- Currently using mean pooling, which amplifies poor effects caused from silence. Will try switching to attention pooling to see if there's an improvement
- Add evaluation by attack type (A07, A09, A13)
  - Currently the EER reflects the overall metric, and breaking down by attack type will help identify where the models are strong/weak

- Error analysis: why were false positives being incorrectly flagged and false negatives categorized are being missed
    - Could look into top 20 worst mistakes